Chapter-7

WORK, ENERGY AND POWER

Work(w):

$W = \vec{F}.\vec{d}$

i) If $\theta < 90°$, w = +ve

ii) If $\theta = 90°$, w = 0

iii) If $\theta > 90°$, w = –ve

It is scalar, unit:– joule, $[w] = [ML^2T^{-2}]$

⇒ Work done by constant force $w = Fd\cos\theta$

*** Work done by variable force**

i) One dimensional variation of force,

$$w = \int_{x_1}^{x_2} F_x d_x$$

ii) Two–dimensional variation,

$$w = \int_{x_1}^{x_2} F_x d_x + \int_{y_1}^{y_2} f_y d_y$$

iii) 3–dimensional variation,

$$w = \int_{x_1}^{x_2} f_x d_x + \int_{y_1}^{y_2} f_y d_y + \int_{z_1}^{z_2} f_z d_z)$$

eg.(i) Find work done by force $\vec{F} = (3x^2\ \vec{i}\ + 2y\ \vec{j}\ )$ N from point (1m, 2m) to (2m, 4m)

$$w = \int_1^2 3x^2dx + \int_2^4 2y\,dy = 19 \text{ Joule}$$

- Work done by spring force either in compression or elongation from natural position.

$$W = -\frac{1}{2}Kx^2$$

where,x = elongation

K = spring constant

or force constant

⇒ Work done by spring force when spring is released either from elongated or compressed position, $\boxed{w = +\frac{1}{2}kx^2}$

⇒ If total work done is round trip is zero then force is said to be **conservative.** Work done by conservative force does not depends on path. i.e., (point function)

Eg.: spring force, gravity, central forces etc.

⇒ If total work done is round trip is not zero then force is said to be non–conservative. Work done by non–conservative force is **path function.**

Energy:–

Kinetic Energy(K):–

$$K = \frac{1}{2}mv^2 = \frac{P^2}{2m}$$

; where P = mv = linear momentum

$\boxed{K \propto P^2}$ $\boxed{\frac{K_2}{K_1} = \left(\frac{P_2}{P_1}\right)^2}$

For small change (< 10%)

$\boxed{\% KE = 2 * \% P}$

For any change

(i) If P is increased by x% then % increase in KE

$\boxed{\% K = (n^2 - 1) \times 100}$

Where, $n = 1 + \frac{\%x}{100}$

(ii) If P is decreased by x% then % decrease in KE

$\% K = (1 - n^2) \times 100 : n = \left(1 - \frac{\%x}{100}\right)$

If linear momentum of body is increased by 50% then % increase in KE will be

$n = 1 + \frac{50}{100} = 1.5,$

$\% k = (1.5^2 - 1) \times 100 = 125\%$

(iii) $\boxed{\% P = (\sqrt{n} - 1) \times 100}$

- If two bodies are released with equal KE then,

$P \propto \sqrt{m}$: i.e., $\boxed{\frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}}}$

- If the linear–momentum of two bodies are equal then,

$K \propto \frac{1}{m}$: i.e., $\boxed{\frac{K_1}{K_2} = \frac{m_2}{m_1}}$

Example:

If bullet is fired from gun then the kinetic energy of gun will be

(a) less then KE of bullet (b) Greater then KE of bullet

(c) equal to KE of bullet (d) none of these

Sol$_n$:- these momentum is constant

$\frac{K_G}{K_b} = \frac{m_b}{m_G} < 1 \Rightarrow \boxed{K_G < K_b}$

- $\boxed{1J = 10^7 \text{ erg}}$ $\boxed{1 \text{ calorie} = 4.2j}$

$\boxed{1eV = 1.6 \times 10^{-19} J}$

- KE is always +ve.

Potential Energy(U):- Energy due to position or configuration.

- Gravitation potential energy of a body of mass 'm' at height 'h' above earth's surface.

ΔU = Change in PE = increase in PE

$$\boxed{\Delta U = mgh\left(\frac{R}{R+h}\right)}$$

(i) If h = nR, then

$$\boxed{\Delta U = mgR * \frac{n}{1+n}}$$

(ii) If h < < < R, then ΔU = mgh

A uniform rod of mass 'm' having length 'ℓ' is lying on horizontal surface if rod is rotated thorng angle 'θ' with horizontal then increase in PE.

$$\boxed{\Delta U = mg\frac{l}{2}\sin\theta}$$

Potential Energy shred in spring (U) = $\frac{1}{2}Kx^2$

(i) If two spring of spring constants K_1 and K_2 are elongated by same force then ratio of PE in it,

$$\boxed{\frac{U_1}{U_2} = \frac{K_2}{K_1}} \quad \boxed{U = \frac{F^2}{2K}}$$

(ii) If elongations in two springs are same then, $\boxed{\frac{U_1}{U_2} = \frac{K_1}{K_2}}$

Work-Energy theorem:

Statement: Work done by all the forces (external and internal both) changes the KE of the body.

$$\boxed{\Sigma w = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2}$$

(i) If KE of body increases then, w = +ve

(ii) If KE of body decrease then, w = –ve

(iii) If KE of body is constant, then, w = 0

- Work done by external force only changes the **total energy** of body.
- For conservative force system, $\boxed{\Delta K + \Delta U = 0}$

i.e., $\boxed{\Delta K = -\Delta U}$

- Loss in KE = Gain in PE and vice-versa

$$\boxed{F = -\frac{dU}{dr}}$$

Power(P):- Rate of doing work.

$$P = \frac{\text{work done}}{\text{time}} = \frac{\vec{F}.\vec{d}}{t}$$

$$\boxed{\vec{F}.\vec{v} = Fv\cos\theta}$$

$$P_{av} = \langle P \rangle = \frac{\Delta w}{\Delta t}$$

$$P_{instant} = \frac{dw}{dt}$$

- If P is the instantaneous power then average power is given by

$$\boxed{\langle P \rangle = \frac{\int_0^t P\,dt}{\int_0^t dt}}$$

Eg. power of a body varying according to equation P = (3t^2 + 2t + 1) watt then power of the body at 2 sec and during 2 sec.

Sol^n. $P\big|_{t=2} = 3 \times 2^2 + 2 \times 2 + 1 = 17$ watt

$$P_{av} = \frac{\int_0^2 (3t^2 + 2t + 1)dt}{\int_0^2 dt} = \frac{14}{2} = 7 \text{ watt}$$

- If a machine delivers constant power to a body then.

(i) velocity of the body, $\boxed{v \propto t^{1/2}}$

(ii) distance moved by body, $\boxed{s \propto t^{3/2}}$

(iii) acceleration of body, $\boxed{a \propto t^{-1/2}}$

(iv) velocity of body, $\boxed{v \propto s^{1/3}}$

- If constant force acts on the body then power of then body.

p $\propto$ t

- If a bullet loses $\frac{1^{th}}{n}$ of its velocity after penetrating through a plank of thickness 't' then

(i) No. of such plank required to stop the bullet

$$N = \frac{n^2}{2n-1}$$

(ii) Total thickness of penetration

$$T = \left(\frac{n^2}{2n-1}\right) \times t$$

eg. (i) If a bullet loses $\frac{1}{20}^{th}$ of its velocity after penetration of plank of thickness '2'cm then,

$$\text{No. of planks (N)} = \frac{20^2}{2 \times 20 - 1} = \frac{400}{39} = 10.256$$

N = 11 Ans

Total thickness of penetration (T) = 10.256 × 2 = 20.51cm

- If a uniform chain/rope of length 'ℓ' and mass 'm' is lying on smooth horizontal table and $\frac{1}{n}^{th}$ of its length $\left(\ell_0 = \frac{\ell}{n}\right)$ is hanging then work done required to put the chain on the table

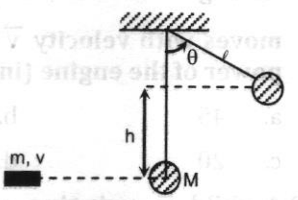
$$W = \frac{mg\ell}{2n^2} = \frac{mg\,\ell_0^2}{2\ell}$$

- If such chain starts slipping from rest then its velocity when completely slip off from the table.

$$V = \sqrt{\frac{g}{\ell}(\ell^2 - \ell_0^2)} = \sqrt{g\ell\left(1 - \frac{1}{n^2}\right)}$$

- If a bullet of mass 'm' moving horizontally with velocity 'v' strikes a block of mass 'm' suspended by a string of length 'ℓ' as shown in figure.

(i) If bullet get embedded in the block then, maximum angle made by string with vertical.



$$\cos\theta = 1 - \frac{m^2v^2}{2g\ell(M+m)}$$

$$\text{Height raised by block (h)} = \frac{m^2v^2}{2g(M+m)^2}$$

Tension in the string T = (M + m) g cosθ

(ii) If bullet emerges out with velocity v_b then maximum angle made by string with vertical.

$$\cos\theta = 1 - \frac{m^2(v - v_b)^2}{2g\ell(M+m)^2}$$

Multiple Choice Questions

1. If a body is placed on another body which is accelerating by an external force then work done by frictional force on the upper body relative to ground is:

a. – ve b. zero

c. + ve d. unity

2. A position dependent force F = 7 – 2x + 3x² newton acts on a small body of mass 2 kg and displaces it from x = 0 to x = 5m. The work done in joule is:

a. 70 b. 270

c. 35 d. 135

3. A body is falling under gravity. When it loses a gravitational potential energy by U, its speed is v. The mass of the body shall be:

a. $\frac{2U}{v}$ b. $\frac{U}{2v}$

c. $\frac{2U}{v^2}$ d. $\frac{U}{2v^2}$

4. Under the action of a force, a 2 kg body moves such that its position x as a function of time is given by $x = \frac{t^3}{3}$, where x is in metres and t in seconds. The work done by the force in the first two seconds is:

a. 1.6 J b. 16 J

c. 160 J d. 1600 J

5. A bock of mass m is pulled along a horizontal surface by applying a force at an angle θ with the horizontal. If the block travels with a uniform velocity and has a displacement d and the coefficient of friction is μ, then the work done by the applied force is:

a. $\frac{\mu mgd}{\cos\theta + \mu\sin\theta}$ b. $\frac{\mu mgd\cos\theta}{\cos\theta + \mu\sin\theta}$

c. $\frac{\mu mgd\sin\theta}{\cos\theta + \mu\sin\theta}$ d. $\frac{\mu mgd\cos\theta}{\cos\theta - \mu\sin\theta}$

6. A mass M is lowered with the help of a string by a distance x at a constant acceleration $\frac{g}{2}$. The work done by the string will be:

a. Mgx b. $\frac{1}{2}Mgx^2$

c. $\frac{1}{2}Mgx$ d. Mgx^2

7. Given that the displacement of the body in metre is a function of time as follows:

$x = 2t^4 + 5$

The mass of the body is 2 kg. What is the increase in its kinetic energy one second after the start of motion?

a. 8 J b. 16 J

c. 32 J d. 64 J

8. The net work done by kinetic friction:

a. is always negative

b. is always zero

c. may be negative and positive

d. is always positive

9. **The work done by pseudo forces is:**
   a. positive b. negative
   c. zero d. all of these

10. **The work done by a spring force:**
   a. is always negative
   b. is always positive
   c. may be negative
   d. may be positive and negative

11. **A particle is moving in a conservative force field from point A to point B. U_A and U_B are the potential energies of the particle at point A and B and W_C is the work done by conservative forces in the process of taking the particle from A to B:**
   a. $W_C = U_B - U_A$ b. $W_C = U_A - U_B$
   c. $U_A > U_B$ d. $U_B > U_A$

12. **A body of mass m thrown vertically upwards attains a maximum height h. At what height will its kinetic energy be 75% of its initial value?**
   a. $\frac{h}{6}$ b. $\frac{h}{5}$
   c. $\frac{h}{4}$ d. $\frac{h}{3}$

13. **A running man has half the K.E. that a body of half his mass has. The man speeds up by $1ms^{-1}$ and then has the same K.E. as that of body. The original speeds of man and body in ms^{-1} are:**
   a. $(\sqrt{2}+1), (\sqrt{2}-1)$
   b. $(\sqrt{2}+1), 2(\sqrt{2}+1)$
   c. $\sqrt{2}, \sqrt{2}$
   d. $(\sqrt{2}+1), 2(\sqrt{2}-1)$

14. **A uniform chain has a mass M and length L. It is placed on a frictionless table with length l_0 hanging over the edge. The chain begins to slide down. Then the speed V with which the end slides away from the edge is given by:**
   a. $V=\sqrt{\frac{g}{L}(L+l_0)}$ b. $V=\sqrt{\frac{g}{L}(L-l_0)}$
   c. $V=\sqrt{\frac{g}{L}(L^2-l_0^2)}$ d. $V=\sqrt{2g(L-l_0)}$

15. **The energy required to accelerate a car from rest to $10\ ms^{-1}$ is W. The energy required to accelerate the car from $10\ ms^{-1}$ to $20\ ms^{-1}$ is:**
   a. W b. 2 W
   c. 3 W d. 4 W

16. **Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. Find the change in its kinetic energy between time t = 2 s and t = 4 s**
   a. 32 J b. 46 J
   c. 61 J d. 102 J

17. **The potential energy between the atoms in a molecule is given by:**
   $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$
   where a and b are positive constants and x is the distance between the atoms. The atom is in stable equilibrium when:
   a. $x = 0$ b. $x = \left(\frac{a}{2b}\right)^{\frac{1}{6}}$
   c. $x = \left(\frac{2a}{b}\right)^{\frac{1}{6}}$ d. $x = \left(\frac{11a}{5b}\right)^{\frac{1}{6}}$

18. **A car moving with a speed of 40 km/h can be stopped by applying brakes after at least 2 m. If the same car is moving with a speed of 80 km/h. What is the minimum stopping distance?**
   a. 8 m b. 2 m
   c. 4 m (d) 6 m

19. **An engine exerts a force $\vec{F} = (20\hat{i} - 3\hat{j} + 5\hat{k})$ N and moves with velocity $\vec{v} = [6\hat{i} + 20\hat{j} - 3\hat{k})$ m/s. The power of the engine (in watt) is:**
   a. 45 b. 75
   c. 20 d. 10

20. **A child is swinging a swing. Minimum and the maximum heights of swing from earth's surface are 0.75 m and 2 m respectively. The maximum velocity of this swing is ($g = 10m/s^2$)**
   a. 5 m/s b. 10 m/s
   c. 15 m/s d. 20 m/s

21. **A stone is thrown at an angle of 45° to the horizontal with kinetic energy K. The kinetic energy at the highest point is:**
   a. $\frac{K}{2}$ b. $\frac{K}{\sqrt{2}}$
   c. K d. zero

22. **When a long spring is stretched by 2cm, potential energy is U. If the spring is stretched 10 cm, the potential energy stored in it will**
   a. 10 U b. 25 U
   c. $\frac{U}{5}$ d. 5U

23. A car of mass m is driven with acceleration a along a straight level road against constant external resistance R. When the velocity is v, the power of the engine is:

a. (R – ma)v b. (R + ma)v

c. mav d. Rv

24. A bread gives a boy of mass 40 kg an energy of 21 kJ. If the efficiency is 28%, then the height can be climbed by him using this energy, is:

a. 22.5 m b. 15 m

c. 10 m d. 5 m

25. The mass of two substances are 25 g and 89 g respectively. If their kinetic energies are same then their ratio of their moments is:

a. 9 : 5 b. 7 : 3

c. 5 : 9 d. none of these

26. Two springs of spring constant 1500 N/m and 3000 N/m respectively are stretched with the same force They will have the potential energies in the ratio of:

a. 1 : 2 b. 1 : 4

c. 4 : 1 d. 2 : 1

27. A simple pendulum of length l has a maximum angular displacement θ. The maximum kinetic energy o the bob of mass m will be:

a. $mgl(1 - \cos\theta)$ b. $mgl\cos\theta$

c. $2\ mgl$ d. mgl

28. Work done by a simple pendulum is one complete oscillation is:

a. $\frac{1}{2\pi}\sqrt{\frac{l}{g}}$ b. $v\sin\frac{\theta}{2}$

c. $\sqrt{mg}$ d. zero

29. A body of mass 5 kg is raised vertically to a height of 10m by a force 170N. The velocity of the body at this height will be

a. 9.8 m/s b. 15 m/s

c. 22 m/s d. 37 m/s

30. A chain of mass M and length *l* is placed on a table with one sixth of it hanging freely from the table edge. The amount of work done to pull the chain on the table is:

a. $\frac{Mgl}{36}$ b. $\frac{Mgl}{72}$

c. $\frac{Mgl}{4}$ d. $\frac{Mgl}{12}$

31. If the force applied is F and the velocity gained in v, then the power developed is:

a. $\frac{v}{F}$ b. Fv

c. Fv^2 d. $\frac{F}{v}$

32. A body of mass 5 kg has momentum of 10 kg m/sec. When a force of 0.2 N is applied on it for 10 sec, the change in its kinetic energy is:

a. 4.4 J (b) 3.3 J

c. 5.5 J d. 1.1 J

33. A pump can hoist 9000 kg of coal per hour from a mine of 120 m deep. Then the power in watts, assuming that its efficiency is 75% is:

a. 4920 watt b. 5920 watt

c. 3920 watt (d) none of these

34. If a man increases his speed by 2 m/s its kinetic energy is doubled. The original speed of the man is:

a. 4 m/s b. $(1 + 2\sqrt{2})$ m/s

c. $(2 + \sqrt{2})$m/s d. $(2 + 2\sqrt{2})$ m/s

35. A ship of mass 2 × 10⁷ kg is at rest. On applying a force of 25 × 10⁵N, it is displaced through 25m. The speed of ship after this displacement will be:

a. 12.5 m/s b. 15 m/s

c. 2.5 m/s d. 3.8 m/s

36. A constant force acts on a mass m at rest. The velocity acquired in traversing a specific distance is directly proportional to:

a. $\frac{1}{\sqrt{m}}$ b. $\sqrt{m}$

c. $\frac{1}{m}$ d. m

37. If the momentum of a certain body be increased by 50%, its kinetic energy will increase by:

a. 125% b. 150%

c. 50% d. 62.5%

38. A motor drives a body along a straight line with a constant force. The power P delivered by the motor must vary with time t according to:

a. t b. $t^{1/2}$

c. $t^{3/2}$ d. t^2

39. What is the velocity of the bob of a simple pendulum at its mean position, if it is able to rise to vertical height of 10cm?

($g = 9.8\ m/s^2$)

a. 2.2 m/s b. 1.8 m/s

c. 1.4 m/s d. 0.6 m/s

40. A force $\vec{F} = (5\hat{i} + 3\hat{j})$ newton is applied over a particle which displaces it from its original to the point $\vec{s} = (2\hat{i} - \hat{j})$. The work done on the particle is:

a. + 11 joule b. + 7 joule
c. + 13 joule d. – 7 joule

41. The potential energy of a body is given by $A - Bx^2$ (where x is the displacement). The magnitude of force acting on the particle is:

a. inversely proportional to x
b. proportional to x^2
c. proportion to x
d. constant

42. A particle is kept at rest at the top of a sphere of diameter 42m. When disturbed slightly, it slides down. At what height h from the bottom, the particle will leave the sphere. [IOE]

a. 35 m b. 16 m
c. 12 m (d) 7 m

43. Calculating the horse power used in pumping 2500 kg of water per minute from a well 12 m deep to the surface. Supposing 40% of the H.P. of the engine during the pumping is wasted. The H.P. of engine will be:

a. 15.6 H.P. b. 6.5 H.P.
c. 11.2 H.P. d. 16.4 J

44. The work done in pulling up a block of wood weighting 2kN for a distance of 10 m on a smooth plane inclined at an angle of 15°, with the horizontal is:

a. 9.82 kJ b. 8.92 kJ
c. 5.18 kJ d. 5.56 kJ

Answers:

1. c	2. d	3. c	4. b	5. b
6. c	7. d	8. c	9. d	10. d
11. b	12. c	13. b	14. c	15. c
16. b	17. c	18. a	19. a	20. a
21. a	22. b	23. b	24. b	25. d
26. d	27. a	28. d	29. c	30. b
31. b	32. a	33. c	34. d	35. c
36. a	37. a	38. a	39. c	40. b
41. c	42. a	43. c	44. c	

Solution for Selected Problems

2. $F = (7 - 2x + 3x^2)$

$$W = \int_0^5 f \cdot dx = [7x - x^2 + x^3]_0^5$$

$$= 35 - 25 + 125$$

$$W = 135\ J$$

3. loss in PE = Gain in KE

$$u = \frac{1}{2}mv^2$$

$$\therefore m = \frac{2u}{v^2}$$

4. $x = \frac{t^3}{3}$

$$v = \frac{dx}{dt} = t^2$$

At t = 0, v = 0

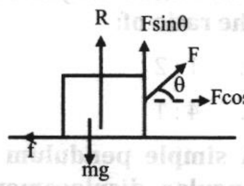
At t = 2sec, v = 4

work done = chang in KE

$$= \frac{1}{2} \times 2 \times 4^2 - \frac{1}{2} \times 2 \times 0^2$$

$$w = 16J$$

5.



$$R = mg - F\sin\theta$$

$$f = \mu R - \mu(mg - F\sin\theta)$$

$$F\cos\theta - f = 0$$

$$F = \frac{\mu\, mg}{\cos\theta + \mu \sin\theta}$$

$$\therefore w = Fd\cos\theta = \frac{\mu\, mg\, d\cos\theta}{\cos\theta + \mu\sin\theta}$$

6. $T = mg - ma = mg - \frac{mg}{2} = \frac{mg}{2}$

work done, $w = T \cdot x = \frac{1}{2} mg\, x$

7. $x = 2t^4 + 5$

$$V = \frac{dx}{dt} = 8t^3$$

At t = 0, V = 0

At t = 1 = 8

theorem in KE $= \frac{1}{2} \times 2 \times 8^2 - 0 = 645$

12. Gain in PE = Loss in KE

$$mgh = \frac{1}{2}mv^2$$

$$\therefore h = \frac{v^2}{2g}$$

Again, Gain in PE = loss in KE

$= \frac{1}{2} mv^2 - 75\%$ of $(\frac{1}{2} mv^2)$

$mgh' = \frac{1}{4} \times \frac{1}{2} mv^2$

$h' = \frac{1}{4} \times \left(\frac{v^2}{2g}\right) = \frac{h}{4}$

13. Let m is the mass of man,

V_m = velocity of man

V_b = velocity of body

$K_{man} = 2 k_{body}$

$\Rightarrow \frac{1}{2} mv_m^2 = \frac{1}{2} \times \frac{1}{2} (m/2) v_b^2$

$\therefore V_m = V_{b/2} \Rightarrow V_b = 2 V_m$

Again,

$\frac{1}{2} m (V_m + 1)^2 = \frac{1}{2} (m/2) v_b^2$

$\Rightarrow (V_m + 1)^2 = 2V_m^2$

On solving:

$V_m = (\sqrt{2} + 1)$

$\therefore V_b = 2(\sqrt{2} + 1)$

15. $\Sigma w = \frac{1}{2} m \times 10^2 - 0$

$\therefore w = \frac{1}{2} m \times 10^2 = \frac{1}{2} m \times 100$

Again, $w' = \frac{1}{2} m [20^2 - 10^2]$

$= \frac{1}{2} m \times 300 = 3w$

16. $\Sigma w = \Delta k$

Again, $P = \frac{dw}{dt}$

$\therefore w = \int_2^4 p.dt] \int_2^4 (3t^2 - 2t + 1)dt$

$w = [t3 - t2 + t]_2^4 = [64 - 16 + 4 - 8 + 4 - 2]$

$\Delta k = 46J$

17. $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$

Force, $F = -\frac{dU}{dx} = -\left[-12 \frac{a}{x^{11}} + \frac{6b}{x^5}\right]$

For equilibrium. F = 0, $x = \left(\frac{2a}{b}\right)^{1/6}$

18. Stopping distance, $S = \frac{V^2}{2a}$

i.e., $S \propto V^2$

$\frac{S_2}{S_1} = \left(\frac{V_2}{V_1}\right)^2$

$\Rightarrow \frac{S_2}{2} = \left(\frac{80}{40}\right)^2$

$\therefore S_2 = 8m$

19. **Power** $P = \vec{F} . \vec{V}$

$= (20\vec{i} - 3\vec{j} + 5\vec{k}).(6\vec{i} - 20\vec{j} - 3\vec{k})$

$= 120 - 60 - 15 = 45$ watt

20. $\frac{1}{2} m V_{max}^2 = mg (h_{max} - h_{min})$

$\therefore V_{max} = \sqrt{2 \times 10 \times (2 - 0.75)}$

$V_{max} = 5m/s$

21. KE at highest point $K = K\cos^2\theta = k\cos^2 45° = k/2$

22. $U = \frac{1}{2} kx^2$

$\frac{U_2}{U_1} = \left(\frac{x_2}{M}\right)^2 = \left(\frac{10}{2}\right)^2$

$\therefore U_2 = 25U$

24. Efficiency $= \frac{output}{Input}$

$\Rightarrow 0.28 = \frac{mgh}{21 \times 10^3}$

$\therefore h = 15m$

25. KE, $K = \frac{P^2}{2m}$

For same KE $P \propto \sqrt{m}$

$\frac{P_1}{P_2} = \frac{\sqrt{m_1}}{m_2} = \sqrt{\frac{25}{89}}$

26. $U = \frac{F^2}{2K} \Rightarrow U \propto \frac{1}{K}$

$\frac{U_1}{U_2} = \frac{K_2}{k_1} = 2:1$

29. $a = \frac{(170 - 5 \times 9.8)}{5} = 24.2$

$V^2 = u^2 + 2ab = 0 + 2 \times 24.2 \times 10 = 484$

$V = 22m/sec$

32. $K = \frac{p^2}{2m}$; $p_i = 10$

Again, $F = \frac{\Delta p}{\Delta t} \Rightarrow \Delta p = 0.2 \times 10 = 2$

$p_f = 10 + 2 = 12$

$\Delta k = \frac{1}{2m}[p_f^2 - p_i^2] = 4.4J$

34. Let initial speed of man = V

$\frac{1}{2}m(V+2)^2 = 2 \times \left(\frac{1}{2}mv^2\right)$

$\Rightarrow (V+2)^2 = 2v^2$

$\therefore V + 2 = \sqrt{2}V$

$\therefore V = \frac{2}{\sqrt{2} - L} = (2 + 2\sqrt{2})$ m/sec

35. $a = \frac{25 \times 10^5}{2 \times 10^7} = 0.125 \text{ m/s}^2$

$V^2 = u^2 + 2as = 0 + 2 \times 0.125 \times 50$

$\therefore V = 2.5$ m/sec

39. $u = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 0.7} = 1.4$m/sec

40. $W = \vec{F} \cdot \vec{d} = 5 \times 2 - 3 \times 1 = +7$ Joule

41. $U = A - Bx^2$

$F = -\frac{dU}{dx} = 2Bx$

$F \propto x$

42. Height from bottom; $h = \frac{5R}{3} = \frac{5 \times 21}{3} = 35$m

43. Power $= \frac{mgh}{t} = \frac{2500 \times 10 \times 12}{60} = 5000$ watt

Power in HP $= \frac{5000}{746} = 6.7$

Efficiency of machine = 60% = 0.6

$x = \frac{\text{output}}{\text{Input}}$

Input $= \frac{6.7}{0.6} = 11.17$ watt $= 11.2$

Power of engine in H.P. = 11.2

44. Workdone along smooth inclined planes

$W = mg\sin\theta \times s$

$= 2 \times \sin 15° \times 10 = 5.18$KJ

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